

(PTFE) Polytetrafluoroethylene

Physical Properties	Metric	English	Comments
Density	2.15 - 2.3 g/cc	0.0777 - 0.0831 lb/in ³	
Apparent Bulk Density	0.36 - 0.91 g/cc	0.013 - 0.0329 lb/in ³	
Water Absorption	0 - 0.03 %	0 - 0.03 %	
Linear Mold Shrinkage	0.01 - 0.058 cm/cm	0.01 - 0.058 in/in	
Mechanical Properties			
Hardness, Rockwell R	58	58	
Hardness, Shore A	98	98	
Hardness, Shore D	50 - 59	50 - 59	
Tensile Strength, Ultimate	10 - 43 MPa	1450 - 6240 psi	
Tensile Strength, Yield	9 - 30 MPa	1310 - 4350 psi	
Elongation at Break	50 - 650 %	50 - 650 %	
Tensile Modulus	0.4 - 1.8 GPa	58 - 261 ksi	
Flexural Modulus	0.5 - 0.7 GPa	72.5 - 102 ksi	
Compressive Yield Strength	10 - 15 MPa	1450 - 2180 psi	
Poisson's Ratio	0.46	0.46	
Shear Strength	5 MPa	725 psi	
Izod Impact, Notched	1.6 J/cm	3 ft-lb/in	
Izod Impact, Notched, Low Temp	0.8 J/cm	1.5 ft-lb/in	
Charpy Impact Unnotched	NB	NB	
Charpy Impact, Notched	0.5 J/cm ²	2.38 ft-lb/in ²	
Tensile Impact Strength	670 kJ/m ²	319 ft-lb/in ²	
Coefficient of Friction	0.06 - 0.1	0.06 - 0.1	
Limiting Pressure Velocity	0.063 MPa-m/sec	1800 psi-ft/min	
Electrical Properties			
Electrical Resistivity	1e+011 - 1e+018 ohm-cm	1e+011 - 1e+018 ohm-cm	
Surface Resistance	1e+011 - 1e+018 ohm	1e+011 - 1e+018 ohm	
Dielectric Constant	2.1	2.1	
Dielectric Constant, Low Frequency	2.1	2.1	
Dielectric Strength	18 - 105 kV/mm	457 - 2670 kV/in	
Dissipation Factor	0.0001 - 0.0003	0.0001 - 0.0003	
Dissipation Factor, Low Frequency	0.0001 - 0.0003	0.0001 - 0.0003	
Arc Resistance	300 sec	300 sec	

Thermal Properties

CTE, linear -100°C	86 $\mu\text{m}/\text{m}\cdot^\circ\text{C}$	47.8 $\mu\text{in}/\text{in}\cdot^\circ\text{F}$
CTE, linear 20°C	79 - 150 $\mu\text{m}/\text{m}\cdot^\circ\text{C}$	43.9 - 83.3 $\mu\text{in}/\text{in}\cdot^\circ\text{F}$
CTE, linear 100°C	100 - 140 $\mu\text{m}/\text{m}\cdot^\circ\text{C}$	55.6 - 77.8 $\mu\text{in}/\text{in}\cdot^\circ\text{F}$
CTE, linear 250°C	170 - 220 $\mu\text{m}/\text{m}\cdot^\circ\text{C}$	94.4 - 122 $\mu\text{in}/\text{in}\cdot^\circ\text{F}$
Specific Heat Capacity	1.2 - 1.4 $\text{J}/\text{g}\cdot^\circ\text{C}$	0.287 - 0.335 $\text{BTU}/\text{lb}\cdot^\circ\text{F}$
Thermal Conductivity	0.24 - 0.35 $\text{W}/\text{m}\cdot\text{K}$	1.67 - 2.43 $\text{BTU}\cdot\text{in}/\text{hr}\cdot\text{ft}^2\cdot^\circ\text{F}$
Melting Point	330 $^\circ\text{C}$	626 $^\circ\text{F}$
Maximum Service Temperature, Air	260 - 290 $^\circ\text{C}$	500 - 554 $^\circ\text{F}$
Deflection Temperature at 0.46 MPa (66 psi)	73 $^\circ\text{C}$	163 $^\circ\text{F}$
Deflection Temperature at 1.8 MPa (264 psi)	45 - 100 $^\circ\text{C}$	113 - 212 $^\circ\text{F}$
Minimum Service Temperature, Air	-200 $^\circ\text{C}$	-328 $^\circ\text{F}$
Flammability, UL94	V-0	V-0
Oxygen Index	95 %	95 %